REAGAN OBADHA

Software Engineer • Cybersecurity Specialist • ML Engineer

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PROFESSIONAL SUMMARY

Results-driven Software Engineer with deep expertise in full-stack development, cybersecurity, and machine learning. Proficient across the entire software development lifecycle — from secure API design and cryptographic systems to CI/CD pipelines, Agile delivery, and ML model deployment. Author of ChainPay, a production-grade fintech platform combining blockchain integrity, AES-256-GCM encryption, AML compliance, and M-Pesa integration. Adept at remote collaboration, meticulous documentation, and delivering high-impact solutions with precision.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, HTML/CSS, Bash

Backend & APIs: FastAPI, Spring Framework, RESTful API design, ASGI/Uvicorn, Node.js

Frontend: React (basics), HTML5, CSS3, Tkinter

Databases: SQLite (WAL/ACID), MySQL, PostgreSQL, schema design, query optimisation

DevOps & CI/CD: Git, GitHub Actions, CI/CD pipelines, Docker (containerisation basics), Agile/Scrum, Kanban

Cybersecurity: Penetration testing, vulnerability assessment, incident response, system auditing, AES-256-GCM encryption, PBKDF2, HMAC, JWT, OWASP Top 10, STRIDE threat modelling, RBAC, rate limiting, account lockout

Machine Learning / AI: Supervised & unsupervised learning, data preprocessing, feature engineering, model evaluation, scikit-learn, pandas, NumPy, neural network fundamentals, AI application development

Blockchain & Fintech: Permissioned blockchain, Merkle trees, Proof-of-Authority, M-Pesa Daraja API, AML engine, double-entry ledger, KYC workflows

Data Entry & Management: Accurate data entry, spreadsheet management, ETL pipelines, data validation, reconciliation

Tools & Practices: IntelliJ IDEA, PyCharm, VS Code, Jira, GitHub, unit testing, TDD, code review

FEATURED PROJECT

ChainPay — Banking-Grade Fintech Platform | Personal Project | *2024 – Present*

A production-oriented, multi-currency mobile money platform integrating blockchain audit capabilities, cryptographic security, AML compliance, and M-Pesa interoperability.

- Designed a custom permissioned blockchain with Proof-of-Authority consensus, SHA-3 Merkle trees, and full chain validation for tamper-evident transaction auditing
- Built a double-entry ledger system with INTEGER minor-unit balance storage, ACID-compliant SQLite WAL transactions, and UUID-based idempotent deduplication
- Implemented banking-grade cryptographic stack: AES-256-GCM authenticated encryption, PBKDF2-HMAC-SHA256 password hashing (100,000 iterations), HMAC-SHA3-256 signing, and JWT HS256 session management
- Developed AML compliance engine with velocity checks, structuring detection (smurfing), sanctions screening, and automated SAR generation
- Integrated Safaricom Daraja API (M-Pesa STK Push) with full state machine, idempotent callbacks, and duplicate receipt detection
- Architected RESTful FastAPI backend with RBAC, sliding-window rate limiting (30 req/min), account lockout, and append-only audit logging
- Built FX engine supporting USD, EUR, KES, NGN, GBP with USD-pivot cross-rate calculation, bid/ask spread, and TTL-based rate caching

- Implemented offline HMAC voucher system for connectivity-limited payment scenarios with anti-double-spend enforcement

Stack: Python · FastAPI · SQLite WAL · SHA-3 · AES-256-GCM · PBKDF2 · JWT · HMAC · Tkinter · Safaricom Daraja API · PyInstaller

EXPERIENCE & ROLES

Freelance Software Engineer & Data Entry Specialist | Remote / Freelance | *2023*

- Delivered backend API services, secure authentication modules, and database-driven web applications for clients across fintech and e-commerce domains
- Performed accurate and high-volume data entry, validation, and ETL pipeline setup for business datasets with zero-error standards
- Conducted security audits and vulnerability assessments, identifying and remediating OWASP Top 10 risks in client web applications
- Applied Agile/Scrum methodology — sprint planning, daily standups, retrospectives — to manage remote project delivery efficiently
- Automated repetitive workflows with Python scripts, reducing manual effort by up to 60% for recurring data management tasks

EDUCATION

Bachelor of Science in Computer Science | Egerton University, Kenya | *2023 – Present*

- Studied at Egerton University with focus on software engineering, algorithms, data structures, computer networks, and database systems
- Relevant coursework: OOP, DBMS, Web Development, Software Engineering, Data Structures & Algorithms, Computer Networks

Technology Skills Training | Modcom Institute of Technology, Nairobi | *Jan – May 2023*

- Intensive hands-on training in software development, data science, cybersecurity, and artificial intelligence
- Gained practical experience with industry-standard development tools, frameworks, and methodologies

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Technology Skills Training Certificate — Modcom Institute of Technology (May 2023)
- Ongoing self-study in Machine Learning, cybersecurity (CEH concepts), and cloud-native development
- Active engagement with open-source projects and fintech architecture research

CORE COMPETENCIES

Remote Collaboration • Technical Communication • Problem-Solving • Attention to Detail • Adaptability • Self-Motivation • Continuous Learning • Agile Delivery • Mentorship

REFERENCES

Available upon request.